

Deutsch Daily

Design & User Guide

A Progressive Web App for daily German vocabulary learning

github.com/ltlfrari/Deutsch-Daily

Created by David E. Ellis (2026).

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Contents

1. Overview

What Deutsch Daily is and how it was built

2. Technical Architecture

PWA structure, data storage, service worker, file layout

3. Content Structure

CEFR levels, lessons, vocab per level, built-in and supplementary packs

4. User Interface

Today screen, History, Settings, navigation, level selector

5. Learning Progression

Lesson flow, block tests, level exams, unlocking, pack switching

6. Test System

Block mini-tests, level exams, optional pack exams, question types

7. Creating New Vocabulary Packs

How to create and add new packs using Claude AI

1. Overview

Deutsch Daily is a Progressive Web App (PWA) for learning German vocabulary, covering all six CEFR levels from A1 (beginner) through to C2 (near-native mastery). The app was created entirely through vibe coding with Claude AI — the initial version was built in approximately 90 minutes in an airport lounge, using Claude and GitHub Pages to serve the app.

All vocabulary, example sentences and grammatical annotations in the built-in lessons were generated by Claude AI, not sourced from an external dataset. This gives the content a consistent pedagogical structure and quality across all levels.

The app is installed on a device as a PWA — it looks and behaves like a native app, works offline once installed, and can send daily reminder notifications. There is no backend, no database and no server beyond GitHub Pages serving static files.

Key Principles

- Daily habit — one lesson a day, tracked with a streak counter.
- Gated progression — lessons unlock in blocks of five; you must pass a test before moving to the next block.
- Consistent structure — every lesson has the same format regardless of level; only vocabulary complexity increases.
- Offline-first — all content is cached locally; the app works without a connection once installed.
- Expandable — supplementary vocabulary packs can be added at any level without changing the app code.

2. Technical Architecture

File Structure

The entire application lives in four files at the root of the GitHub repository:

File	Purpose
index.html	The complete application — HTML, CSS and JavaScript in a single file (~2,100 lines)
sw.js	Service worker — handles daily push notification scheduling (81 lines)
icon-192.png	PWA home screen icon (192×192 px) — German flag with fabric wave effect
icon-512.png	PWA splash screen icon (512×512 px) — same design at higher resolution

Supplementary vocabulary packs are stored in a `data/` subdirectory:

Path	Purpose
data/manifest.json	Lists all available supplementary packs per CEFR level
data/A1/pack-2.json	Supplementary lesson pack — 25 A1 lessons
data/A1/pack-3.json	Supplementary lesson pack — 25 A1 lessons
data/A1/pack-4.json	Supplementary lesson pack — 25 A1 lessons
data/A2/pack-2.json	Supplementary lesson pack — 25 A2 lessons

State Management

All user progress is stored in the browser's `localStorage` under the key `deutschDaily_v2` as a JSON object. This includes:

- Current level and active pack
- Lesson position within each pack (remembered separately per pack)
- Completion history — last 120 lessons marked as studied, keyed by level, pack and date
- Tests passed — which block mini-tests have been passed, keyed by level and pack
- Final exams passed — which level exams and optional pack exams have been passed
- Streak count and last study date
- Settings — notification time, hide-translations toggle
- Downloaded pack content — pack JSON cached locally so it works offline

Service Worker

The service worker (`sw.js`) handles daily push notifications. When the user grants notification permission and sets a reminder time, the app sends a message to the service worker which schedules a `setTimeout` loop. It fires a notification at the chosen time each day and then re-schedules itself for 24 hours later.

3. Content Structure

CEFR Levels

The app covers all six Common European Framework of Reference for Languages (CEFR) levels. Each level contains exactly 25 lessons grouped into 5 blocks of 5. Levels are unlocked sequentially by passing the previous level's final exam.

Level	Description	Nouns/lesson	Verbs/lesson	Adjs/lesson	Sentences/lesson
A1	Beginner	2	1	1	2
A2	Elementary	3	2	2	2
B1	Intermediate	4	3	3	2
B2	Upper-Intermediate	5	4	4	2
C1	Advanced	5	4	4	2
C2	Mastery	5	4	4	2

Note: C1 and C2 have the same vocabulary count per lesson as B2. The increase in difficulty comes from the register and specialisation of the vocabulary — C1 covers academic and professional language, C2 covers near-native mastery with highly specialised terms.

Lesson Format

Every lesson, at every level, follows the same structure:

- **Nouns** — each with the definite article (der/die/das), English translation, and grammatical gender note (masculine, feminine, or neuter).
- **Verbs** — infinitive form, English translation, and a conjugation example in the er/sie/es form.
- **Adjectives** — German word and English translation.
- **Sentences** — two bilingual example sentences (German and English) that use the lesson vocabulary in context.

Vocabulary Packs

Each CEFR level has a built-in pack of 25 lessons. Once the built-in pack is completed and the level exam passed, supplementary packs become available. Each supplementary pack is identical in structure to the built-in pack: 25 lessons, 5 blocks of 5, with the same lesson format.

Currently available supplementary packs:

- A1: Pack 2, Pack 3, Pack 4 (75 additional lessons, 300+ extra vocabulary items)
- A2: Pack 2 (25 additional lessons, 75+ extra vocabulary items)
- B1 through C2: built-in pack only (supplementary packs can be added at any time)

4. User Interface

The app has three main screens accessed via a bottom navigation bar: Today, History and Settings.

Header

The header shows "Deutsch Daily" on the left and a level selector on the right. The level selector displays the current level highlighted in gold, with any previously unlocked levels shown as tappable pills. Tapping an unlocked level switches to it immediately, resuming at your last position in that level.

Today Screen

The main screen where all learning takes place. From top to bottom:

- **Streak and done-today badge** — your current daily study streak and how many lessons you have marked as studied today.
- **Lesson navigation** — left (<) and right (>) arrows with a counter showing "Lesson N of 25". A coloured badge appears in the counter when a test is ready or needed.
- **Pack switcher** — once the built-in level exam is passed, a row of pills appears below the counter showing Built-in and any available supplementary packs. The active pack is highlighted. Tap any pill to switch packs; your position in each pack is remembered.
- **Lesson content** — the vocabulary cards for the current lesson, colour-coded by type: gold for nouns (showing article, German word, gender note and English), green for verbs (showing infinitive, conjugation and English), red for adjectives. Example sentences appear at the bottom.
- **Hide translations** — if enabled in Settings, translations are hidden behind a "Reveal translation" button on each card.
- **Mark as studied** — a button at the bottom of each lesson. Tapping it records the lesson as complete for today and updates the streak.
- **Test/exam banners** — when all five lessons in a block are marked complete, a "Test N ready!" badge appears in the counter. When all five block tests are passed, a "Level Exam Ready" banner appears above the lesson content.

History Screen

Shows a reverse-chronological log of all lessons studied, grouped by date. Each entry shows the date, the vocabulary covered in that lesson as small chips, and which block tests and level exams were completed on that date. Completed tests show a "Retake" button for practice retakes.

Settings Screen

Contains three settings:

- **Level selector** — six CEFR level pills (A1 through C2). Locked levels are greyed out. Selecting a level switches the app to that level. Note: the header level selector is faster for switching between already-unlocked levels.

- **Daily reminder** — a toggle to enable push notifications with a time picker. Requires notification permission to be granted in the browser.
- **Hide translations** — when on, English translations on lesson cards are hidden and must be revealed individually. Useful for self-testing.

5. Learning Progression

Overview

Progression through the app is gated — you cannot skip ahead. Within a level, lessons unlock in blocks of five. You work through blocks sequentially, taking a test after each block before the next five lessons unlock.

Step-by-Step Flow

Start

Open the app. You begin on A1, Lesson 1.

Study lessons 1-5

Navigate through the five lessons using the < > arrows. Read the vocabulary and example sentences. Mark each lesson as studied using the button at the bottom. You can do lessons in any order within the block.

Complete Block 1

Once all five lessons are marked as studied, a "■ Test 1 ready!" badge appears in the lesson counter, regardless of which lesson you are currently viewing.

Take Block Test 1

Tap the "Test 1 ready!" badge. A 12-question test covering the five lessons appears. Answer all questions. You need 11 or 12 correct (90%) to pass.

Pass or retry

If you pass, lessons 6-10 unlock immediately. If you fail, you return to the lessons to review and can retake the test. There is no limit on retakes.

Continue through all blocks

Repeat for blocks 2-5 (lessons 6-10, 11-15, 16-20, 21-25). Each block requires all five lessons to be studied before the test becomes available.

Level Exam

After passing all five block tests, a "Level Exam Ready!" banner appears on the Today screen. Tap "Take Level Exam" to start the 40-question final exam covering all 25 lessons in the level.

Pass the Level Exam

Score 36/40 or more (90%) to pass. On passing, you are offered two choices: advance to the next CEFR level, or stay on the current level and do a supplementary vocabulary pack.

Supplementary packs

If you stay, the pack switcher appears. Download a supplementary pack and work through it the same way — 25 lessons, 5 block tests. The final exam for a supplementary pack is optional (no gating); it is offered as a self-assessment tool.

Next level

If you advance, the app switches to the next CEFR level and the cycle begins again. You can return to a previous level at any time using the header level selector.

Navigation Rules

- You can freely navigate backwards through lessons at any time.
- You can navigate forward freely within the current unlocked block.
- Pressing › on the last lesson of a block when the test has not been passed shows an alert if lessons are incomplete, or a "■ Test needed" badge if all lessons are done. Tap the badge to go to the test.
- You cannot scroll past the end of the last unlocked block until the test is passed.
- When the level exam banner is showing, pressing › on lesson 25 does nothing — the only way forward is to tap the Level Exam button.

6. Test System

Block Mini-Tests

After completing every block of five lessons a 12-question test must be passed before the next block unlocks. The questions are drawn from the five lessons in that block. A score of 90% or better (11 or 12 correct) is required to pass.

Question Type	Count	Description
Noun translation	3	Choose the correct English translation for a German noun
Verb translation	2	Choose the correct English translation for a German verb
Adjective translation	1	Choose the correct English translation for a German adjective
Sentence translation	2	Choose the correct English translation for a German sentence
Article selection	4	Choose the correct article (der / die / das) for a noun
Total	12	90% pass rate — 11 or 12 correct required

Level Final Exam

After passing all five block tests in a level, the 40-question final exam becomes available. This exam draws questions from all 25 lessons across the entire level. A score of 90% or better (36 or more correct) is required to unlock the next CEFR level.

Question Type	Count
Noun translation	14
Verb translation	8
Adjective translation	6
Sentence translation	6
Article selection	6
Total	40

Optional Pack Exam

Supplementary packs have an optional 40-question exam offered after all 25 lessons and 5 block tests are completed. This exam uses the same format as the level final exam but the result has no gating effect — it cannot lock or unlock anything. It is purely a self-assessment tool. If passed, the pack pill in the switcher row shows a ✓ tick.

Test Behaviour

- Questions are presented one at a time with four multiple-choice options.
- Distractors (wrong answers) are drawn from other vocabulary in the same block or level.

- Immediate feedback is shown after each answer — correct answers turn green, wrong answers turn red and show the correct answer.
- A progress bar and score counter run throughout the test.
- Passed block tests can be retaken in practice mode from the History screen — the retake does not affect the pass record.
- There is no time limit on any test.

7. Creating New Vocabulary Packs

New vocabulary packs are JSON files containing 25 lessons each, stored in the repository under `data/[LEVEL]/` and listed in `data/manifest.json`. Once pushed to GitHub, packs appear automatically in the app's pack switcher for any user who has completed the built-in level.

The recommended method is to ask Claude to do the entire job — reading the existing packs from GitHub, analysing them for used vocabulary, generating new content, and pushing everything back. The manual method is also described as an alternative.

Pack File Format

Each pack is a JSON array of exactly 25 lesson objects. The number of vocabulary items per lesson must match the target CEFR level:

Level	Nouns	Verbs	Adjectives	Sentences
A1	2	1	1	2
A2	3	2	2	2
B1	4	3	3	2
B2 / C1 / C2	5	4	4	2

Example A1 lesson object:

```
[ { "nouns": [ { "de": "die Blume", "en": "the flower", "note": "feminine"}, { "de": "der Garten", "en": "the garden", "note": "masculine"} ], "verbs": [ { "de": "spielen", "en": "to play", "conj": "er spielt"} ], "adjs": [ { "de": "schoen", "en": "beautiful"} ], "sentences": [ { "de": "Die Blume im Garten ist schoen.", "en": "The flower in the garden is beautiful."}, { "de": "Das Kind spielt im Garten.", "en": "The child plays in the garden."} ] }, ... (25 lessons total) ]
```

Method 1 — Ask Claude (Recommended)

This is by far the easiest approach. Claude can read the existing packs from GitHub, work out which words have already been used, generate the new content and push everything back to the repository in a single conversation. No manual file handling is required.

What you need before starting:

- A GitHub personal access token with write access to the repository.
- A conversation with Claude (claude.ai) — this works best in the same session as the development work, or in a new session where Claude has access to the repository.

What to say to Claude:

Simply tell Claude what you want. For example:

```
"I need two new A1 vocabulary packs and one new B1 pack for Deutsch Daily. Clone the repo at github.com/ltlfrari/Deutsch-Daily, read all the existing A1 and B1 packs and
```

```
the built-in lessons in index.html, then generate the new packs avoiding any vocabulary already used. Update the manifest and push everything. My token is ghp_xxxxxxxxxxxxx"
```

Claude will then:

- Clone the repository and read all existing pack files for the requested levels.
- Extract every German word (nouns, verbs, adjectives) already in use across the built-in lessons and all supplementary packs.
- Generate the requested number of new packs, each with exactly 25 lessons, using only fresh vocabulary appropriate for the target level.
- Save the new pack files in the correct directory (e.g. data/A1/pack-5.json).
- Update data/manifest.json to include the new pack IDs.
- Commit and push everything to the repository. The packs are live immediately.

Note: Remember to regenerate your GitHub token after sharing it in a conversation. Tokens shared in chat should be rotated once used.

Method 2 — Manual Process

If you prefer to create packs yourself without Claude doing the GitHub work, follow these steps.

Step 1 — Get the used-words list

Before generating new content, compile a list of all German words already used at the target level. Ask Claude in a new conversation:

```
"Read the A1 section of index.html at github.com/ltlfrari/Deutsch-Daily and all the files in data/A1/. List every German noun, verb and adjective used, as a comma-separated list."
```

Step 2 — Generate the pack

Ask Claude to generate the new pack content, passing the used-words list to avoid repetition:

```
"Generate a complete A1 German vocabulary pack as a JSON array of exactly 25 lessons. Do not use any of these words: [paste list here]. Rules: - 2 nouns, 1 verb, 1 adjective, 2 sentences per lesson - Each noun: {"de": "die X", "en": "the x", "note": "feminine/masculine/neuter"} - Each verb: {"de": "infinitive", "en": "to ...", "conj": "er/sie/es ..."} - Each adjective: {"de": "word", "en": "translation"} - Each sentence: {"de": "German.", "en": "English."} - Sentences should use lesson vocabulary in context - Return ONLY the JSON array, no explanation, no markdown"
```

Step 3 — Validate the JSON

Paste the output into a JSON validator (e.g. jsonlint.com). Confirm it contains exactly 25 lesson objects and that each has the correct number of vocabulary items for the level.

Step 4 — Save the file

Save as pack-N.json where N is the next available number for that level. Place in data/[LEVEL]/ (e.g. data/A1/pack-5.json).

Step 5 — Update the manifest

Open `data/manifest.json` and add the new pack ID to the array for the relevant level:

```
{ "version": 1, "packs": { "A1": ["pack-2", "pack-3", "pack-4", "pack-5"], "A2":  
["pack-2"], "B1": [], ... } }
```

Step 6 — Commit and push

Commit both the new pack file and the updated manifest to the repository and push to the main branch. The new pack will appear in the app's pack switcher automatically once the manifest is fetched.

Note: The app has no mechanism to detect duplicate vocabulary across packs automatically. Whether using Method 1 or Method 2, always check existing packs before generating new content to avoid repetition.